Fizika

9 sinf

MAVZU: Masalalar yechish

1-namunaviy masala

$3 \cdot 10^{-5} m^3$ hajmli simobli lampa ballonida 10^{12} ta molekula bo‘lsa, undagi simob bug‘lari 300 K temperaturada qanday bosim (μPa) hosil qiladi?

Bolsman doimiysi $1{,}38 \cdot 10^{-23} J/K$.

Berilgan:

$V = 3 \cdot 10^{-5} m^3$

$N = 10^{12} ta$

$\mathrm{T} = 300\ \mathrm{K}$

$k = 1{,}38 \cdot 10^{-23} J/K$

Topish kerak: $p - ?$

Formula:

$$p = n \cdot k \cdot T$$

$$n = \frac{N}{V}$$

$$p = \frac{N}{V} \cdot k \cdot T$$

Yechish:

$$p = \frac{10^{12}}{3 \cdot 10^{-5} m^3} \cdot 1{,}38 \cdot 10^{-23} \frac{J}{K} \cdot 300\ K = 138 \cdot 10^{-6}\ \mathrm{Pa}$$

Javob: $p = 138 \cdot 10^{-6}$ **Pa = 138** μ**Pa**

2-namunaviy masala

Temperaturasi $367℃$ va bosimi $8{,}31 \cdot 10^5$ Pa bo‘lgan kislorodning zichligini aniqlang.

Berilgan:

$t = 367℃$

$p = 8{,}31 \cdot 10^5 Pa$

$R = 8{,}31 \ \frac{J}{K \cdot mol}$

$M_{(O_2)} = 32 \frac{g}{mol} = 32 \cdot 10^{-3} \frac{kg}{mol}$

Topish kerak: $\rho - ?$

Formula:

$\rho = \frac{m}{V}$; $m = m_0 \cdot N$

$m_0 = \frac{M}{N_A}$; $\rho = \frac{M \cdot N}{N_A \cdot V}$

$p = n \cdot k \cdot T$; $n = \frac{N}{V}$

$p = \frac{N}{V} \cdot k \cdot T$

$V = \frac{N \cdot k \cdot T}{p} \Rightarrow \rho = \frac{M \cdot N}{N_A \cdot V} = \frac{M \cdot N \cdot p}{N_A \cdot N \cdot k \cdot T} = \frac{M \cdot p}{T \cdot R}$; $N_A \cdot k = R$

T=t+273

Yechish:

$$\mathrm{T} = 273 + 367°\mathrm{C} = 640\ \mathrm{K}$$

$$\rho = \frac{32 \cdot 10^{-3}\ \frac{\mathrm{kg}}{\mathrm{mol}} \cdot 8{,}31 \cdot 10^{5}\mathrm{Pa}}{640\ \mathrm{K} \cdot 8{,}31\ \frac{\mathrm{J}}{\mathrm{K} \cdot \mathrm{mol}}} = 5\ \mathrm{kg}/m^3$$

Javob: $\boldsymbol{\rho = 5\ kg/m^3}$

3-namunaviy masala

3 g vodoroddagi molekulalar soni **9 g** suvdagiga qaraganda necha marta ko‘p?

Berilgan:

$m_{(H_2)} = 3\ g$

$m_{(H_2O)} = 9\ g$

$M_{(H_2)} = 2\ \frac{g}{mol}$

$M_{(H_2O)} = 18\ \frac{g}{mol}$

T.K.: $\frac{N_{(H_2)}}{N_{(H_2O)}} - ?$

Formula:

$N = \nu \cdot N_A$

$\frac{N_{(H_2)}}{N_{(H_2O)}} = \frac{\nu_{(H_2)} \cdot N_A}{\nu_{(H_2O)} \cdot N_A} = \frac{\nu_{(H_2)}}{\nu_{(H_2O)}}$; $\nu = \frac{m}{M}$

$\frac{N_{(H_2)}}{N_{(H_2O)}} = \frac{m_{(H_2)}}{M_{(H_2)}} : \frac{m_{(H_2O)}}{M_{(H_2O)}} = \frac{m_{(H_2)}}{M_{(H_2)}} \cdot \frac{M_{(H_2O)}}{m_{(H_2O)}}$

Yechish: $\frac{N_{(H_2)}}{N_{(H_2O)}} = \frac{3\ g}{2\ \frac{g}{mol}} \cdot \frac{18\ \frac{g}{mol}}{9\ g} = 3$

Javob: 3 marta katta

4-namunaviy masala

Agar idishdagi havo $0{,}83\ \mu Pa$ bosimgacha so‘rib olingan bo‘lsa, 27℃ temperaturada idishning $1\ mm^3$ hajmida qancha molekulasi bo‘ladi?

Universal gaz doimiysi 8,3 J/K·mol.

Avagadro soni $6 \cdot 10^{-23} mol^{-1}$

Berilgan:

$p = 0{,}83\mu Pa = 8{,}3 \cdot 10^{-7}\ Pa$

$\mathrm{T} = 27℃ + 273 = 300\ \mathrm{K}$

$V = 1\ mm^3 = 10^{-9} m^3$

$R = 8{,}3\ \frac{J}{K \cdot mol}$

$N_A = 6 \cdot 10^{23} mol^{-1}$

Topish kerak: $N-?$

Formula:

$p = n \cdot k \cdot T$

$n = \frac{N}{V} = \frac{p}{k \cdot T}$

$N = \frac{p \cdot V}{k \cdot T}$

$R = k \cdot N_A \Rightarrow k = \frac{R}{N_A}$

$N = \frac{p \cdot V \cdot N_A}{R \cdot T}$

Yechish: $N = \frac{8{,}3 \cdot 10^{-7}\ Pa \cdot 10^{-9} m^3 \cdot 6 \cdot 10^{23} mol^{-1}}{8{,}3 \frac{J}{K \cdot mol} \cdot 300\ K} = 2 \cdot 10^5 ta$

Javob: $2 \cdot 10^5 ta$ molekula bo‘ladi.

5-namunaviy masala

2 mol ideal gazning 27°C temperaturadagi to‘liq ilgarilanma harakat kinetik energiyasi qanday bo‘ladi?

$$R = 8,3\ J/K \cdot mol.$$

Berilgan:

$\mathrm{T} = 27°\mathrm{C} + 273 = 300\ \mathrm{K}$

$\nu = 2\ mol$

$R = 8{,}3\ \frac{J}{K \cdot mol}$

Topish kerak: $E_k - ?$

Formula:

$$E_k = \frac{3}{2}\ N\ k\ T$$

$$N = \nu\ N_A$$

$$E_k = \frac{3}{2}\nu\ N_A\ k\ T; \qquad N_A\ k = R$$

$$E_k = \frac{3}{2}\ \nu\ R\ T$$

Yechish: $E_k = \frac{3}{2} \cdot 2\ mol \cdot 8{,}3\ \frac{J}{K \cdot mol} \cdot 300\ \mathrm{K} = 7470\ \mathrm{J}$

Javob: $E_k = 7470\ \mathrm{J}$

6-namunaviy masala

Xonada elektrokamin yoqilganidan so‘ng o‘zgarmas bosim holatida havo temperaturasi 18℃ dan 27℃ gacha ko‘tarildi.

Xona ichidagi havo molekulalarining soni necha foizga kamaygan?

Berilgan:

$T_1 = 18℃ + 273 = 291\ \text{K}$

$T_2 = 27℃ + 273 = 300\ \text{K}$

$p_1 = p_2$

$V_1 = V_2 = V$

T.K.: $\frac{N_1 - N_2}{N_1} \cdot 100\ \% - ?$

Formula:

$p = n \cdot k \cdot T$

$p_1 = n_1 \cdot k \cdot T_1$

$p_2 = n_2 \cdot k \cdot T_2$

$p_1 = p_2$ dan

$n_1 \cdot k \cdot T_1 = n_2 \cdot k \cdot T_2 \ ; \ n = \frac{N}{V}$

$\frac{N_1}{V} \cdot T_1 = \frac{N_2}{V} \cdot T_2 \Rightarrow N_1 \cdot T_1 = N_2 \cdot T_2$

$$N_2 = N_1 \cdot \frac{T_1}{T_2} \Rightarrow \frac{N_1 - N_2}{N_1} \cdot 100\ \% = \frac{N_1 - N_1 \cdot \frac{T_1}{T_2}}{N_1} \cdot 100\ \% =$$

$$= \left(1 - \frac{T_1}{T_2}\right) \cdot 100\ \%$$

Yechish:

$$\frac{N_1 - N_2}{N_1} \cdot 100\ \% = \left(1 - \frac{291\ K}{300\ K}\right) \cdot 100\ \% = 3\ \%$$

Javob: 3 % ga kamaygan.

Mustaqil bajarish uchun topshiriqlar

1. (44-bet 7-test) Berk idish ichidagi ideal gaz molekulalarining o‘rtacha kvadratik tezligi 30% ga ortsa, gaz bosimi qanday o‘zgarishini toping.

2. (44-bet 10-test) Ballondagi kislorod tempera-turasi $27℃$ dan $227℃$ gacha ko‘tarilsa, gaz zichligi qanday o‘zgaradi?

3. $5 \cdot 10^{24}$ ta gaz molekulasi 300K temperatura va 414 Pa bosimda qanday hajmni egallaydi? Bolsman doimiysi $1{,}38 \cdot 10^{-23}\ J/K$.